Hall Ticket Number:

MR-22

Code: B424CS2S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2024/January 2025

Discrete Mathematics

(Common to CSE, IT and CSM)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

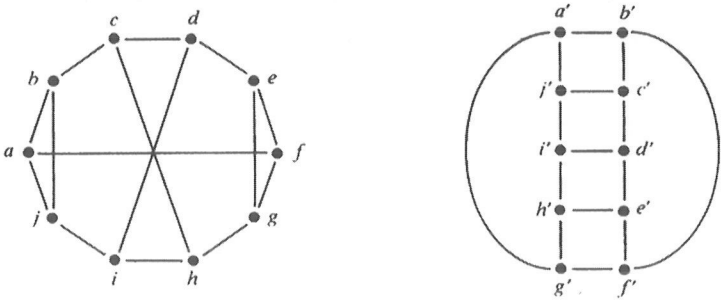
Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define well-formed formula.	1	1	1	1
1. b)	What is Tautology?	1	1	1	1
1. c)	Discuss inclusion and Equality of sets.	1	3	2	1
1. d)	Explain about storage structures.	1	2	2	1
1. e)	Define Lattice with an example.	1	1	3	1
1. f)	Find LUB and GLB for the subset {{a,b},{a},{b}} of power set of A. where A={a,b,c}.	1	3	3	1
1. g)	Use the multinomial theorem to expand $(x_1+x_2+x_3+x_4)^4$.	1	3	4	1
1. h)	Explain permutation and combination with an example.	1	2	4	1
1. i)	Is there a simple graph with degree sequence (1,1,3,3,3,4,6,7)?	1	3	5	1
1. j)	Define spanning tree with example.	1	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Show that SVR is tautologically implied by (PVQ)Λ(P→R)Λ(Q→S).	5	2	1	1,2
2. b)	Obtain PDNF of the formula (¬PV¬Q)→(P↔¬Q).	5	4	1	1,3
	OR				
2. c)	Construct the truth table of the formula (P↔Q)↔((PΛQ)V(¬PΛ¬R)).	5	3	1	1
2. d)	What is predicate, free and bound variables and quantifiers?	5	1	1	1
3. a)	Let f: R→R and g: R→R, where R is the set of real numbers. Find fog and gof where $f(x)=x^2-2$ and g(x)=x+4.State whether these functions are bijective.	5	2	2	1,2
3. b)	Let X={ball, bed, dog, let, egg}, and R is a relation defined on X as R={(x,y)| xRy if x and y contain some common letter},Show that R is compatibility relation and also find maximum compatibility blocks for R.	5	3	2	1
	OR				
3. c)	Let X={1,2,3,4} and R={(1,1),(1,4),(4,1),(4,4),(2,2),(2,3),(3,2),(3,3)} check whether the relation is an equivalence relation, Write the matrix of R and sketch its graph.	5	4	2	1,2
3. d)	Let A ={a,b,c} be a finite set and p(A) its power set. Let ⊆ be the inclusion relation on the elements of p(A).Draw Hasse diagram of (p(A), ⊆).	5	4	2	1
4. a)	Explain about the following. i) Semi group ii) Sub monoid iii) Homomorphism.	5	2	3	1
4. b)	Let S=NxN Let * be the operation on S defined by (a, b) * (c, d) = (ac, bd).show that S is a semigroup under *.	5	3	3	1
	OR				
4. c)	The set A={2,4,5,6,8,9} on which the operation * is defined by a*b=Minimum of b. Find whether the given set forms a semi group or monoid.	5	2	3	1
4. d)	In a lattice (L,≤,Λ,V)state and prove the laws idempotent, commutative, associative and absorption.	5	2	3	1

P.T.O.

5. a)	In how many ways can 14 people be partitioned into 6 teams when the first and second teams have 3 members each and the third, fourth, fifth, and sixth teams have 2 members each?	5	3	4	1
5. b)	Prove that C(n, r) = C(n -l, r) + C(n -l, r- 1).	5	2	4	1
	OR				
5. c)	Find the coefficient of v2xy3z4 In $(v + w + x + y + z)^{10}$.	5	2	4	1,3
5. d)	How many integers between 1 and 1000 are not divisible by 2,3,5 or 7?	5	3	4	1
6. a)	Show that the two graphs are isomorphic or not. 	5	2	5	1,2
6. b)	Show that tree with n vertices has exactly n – 1 edges.	5	2	5	1
	OR				
6. c)	What is a Hamiltonian Cycle? Draw bipartite graph K3,4 and prove that this graph does not have a Hamiltonian cycle.	5	2	5	1
6. d)	Write short notes on DFS and BFS	5	2	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome